

1 What Accountants Need to Know

Introduction

Artificial Intelligence (AI) is transforming how we live and work. Intelligent virtual assistants on smartphones (e.g., Siri on Apple devices), facial recognition software on social media platforms (e.g., Facebook), and self-driving autonomous vehicles (e.g., Tesla) all utilize AI. Hospitals used AI in the battle against Covid-19. For example, Tampa General Hospital in Florida has deployed an AI system that performs facial thermal scans on patients entering the building to detect potential coronavirus symptoms such as sweat, discoloration, and fever (Wittbold et al., 2020). The convergence of big data analytics, advances in computing power, the Internet of Things (IoT), and large scale investment by governments, universities, and high-tech organizations such as Google, Amazon, Microsoft, and IBM have accelerated the adoption of AI for consumers and businesses.

These cases are just a few examples of the digital transformation we are currently experiencing, an era described by Klaus Schwab, founder and executive chairman of the World Economic Forum, as “The Fourth Industrial Revolution”. Schwab notes that the current revolution is distinct from the previous three due to the velocity of technological change, the breadth and depth of these technologies, and its impact on entire systems (i.e., countries, companies, industries, and society as a whole) (Schwab, 2017). AI is one of the key technologies driving this digital revolution.

So, what exactly is AI? **Artificial intelligence** is a computer program or software application that can imitate or simulate human behavior. AI applications are expected to automate a significant portion of the repetitive tasks performed by accountants, though the implications of this change for the accounting profession has divided scholars. Some

2 *What Accountants Need to Know*

critics speculate that sufficiently sophisticated AI will eliminate the need for accountants in the future. Others predict that the accounting profession will experience increased productivity and cost savings by incorporating AI technology. What is evident on both sides is that AI will have a significant impact on the accounting profession in the coming decades. As we will discuss throughout this book, and in more detail in Chapter 7, our view is that AI will not eliminate the need for accountants but empower them to deliver more accurate, timely, and forward-looking insights into the business(es). As part of the impending transformation of the profession, accountants will need to develop the skills necessary to provide mission-critical judgment and oversight of AI-enabled systems and processes in order for organizations to realize productivity gains and meet other objectives expected from AI. The purpose of this book is to provide accountants with a practical guide for identifying opportunities to implement artificial intelligence (AI) initiatives to increase productivity and profitability.

History of AI

Given the recent widespread media coverage on AI, the field of AI may seem like a new discipline. However, the roots of AI's can be traced back to 1943 when neurophysiologist Warren McCulloch and logic expert Walter Pitts proposed the first model of artificial neurons (Russell & Norvig, 2010). The field of AI was established as an academic and research discipline when the first AI conference was organized at Dartmouth College in 1956 (Bringsjord & Govindarajulu, 2018). Today, AI is an interdisciplinary field of science, engineering, statistics, philosophy, neuroscience, psychology, computer engineering, and others.

History of Accountants Using Technology

Technology has always played an important role in accounting for business transactions through the ages – from the use of an abacus over 2,000 years ago (New York State Society of CPAs, 2015), to the purchase of a computer in 1955 exclusively for accounting purposes (Untapped Editorial Team, 2018), to the creation of electronic spreadsheets in the 1980s (Pepe, 2011). AI is the next significant technological advancement that will play an essential role in the accounting profession for years to come.

Overview of How Accountants Are Using AI

Accountants currently use AI in a variety of applications, from identifying high-risk transactions in audits to performing accounts payable processing tasks. As a result, AI is increasingly taking the place of human processing and decision-making, which will continue to have a transformative impact on business practices. Although AI adoption in the accounting profession is in its early stages, practitioners must understand the implications of this technology.

The adoption of AI has been on a steady rise in both public and corporate accounting. For example, EY¹ uses AI drones to perform inventory counts with increased accuracy and efficiency. Deloitte² uses a tool called “Argus” to extract critical accounting information from any type of electronic document to improve the quality and efficiency of an audit (Deloitte, 2015). In management accounting, AI is used to automatically code accounting entries, forecast revenues, and analyze unstructured data such as contracts and emails (ICAEW, 2017).

Going forward, it is even more exciting to think about how AI will be used in accounting. Imagine if you could ask an intelligent virtual assistant (such as Alexa or Siri) to analyze accounts payable for duplicate payments as part of your audit procedures. Technological advances in artificial intelligence could make this possible very soon. Researchers Burns and Igou (2019) suggest that accountants should consider using intelligent virtual assistants to interface with AI applications to perform audit analysis, data retrieval, spreadsheet creation, and data visualizations – all activated by human commands.

Human Intelligence versus Artificial Intelligence

For more than 2,000 years, philosophers have pondered questions such as “how does the human mind work?” and “can non-humans have minds?” (Negnevitsky, 2011, p. 1). The Merriam-Webster dictionary (2020) defines intelligence as follows:

- 1 The ability to learn or understand or to deal with new or trying situations
- 2 The ability to apply knowledge to manipulate one’s environment or to abstractly think as measured by objective criteria (such as tests)

For decades, scholars in the field of psychology have debated what constitutes human intelligence (Weiner & Freedheim, 2013). Some

4 *What Accountants Need to Know*

experts describe human intelligence as the capacity for “reasoning, problem solving and learning” (Colom et al., 2010). Additional common characteristics of human intelligence include logical thinking, spatial perception, and pattern recognition (Yao et al., 2018). Human intelligence also involves the ability to perceive, understand, and predict (Russell & Norvig, 2010) as well as the capacity for planning and adaptability (Siegel et al., 2003).

The goal of AI is to build machines that “can perform complex tasks as well as, or better than, humans. In order to perform those complex tasks, machines must be able to perceive, reason, learn, and communicate” (Siegel et al., 2003, p. 1). Achieving this goal has been the quest of computer scientists and engineers for more than half a decade. Computers are very good at performing specific tasks, such as complex math calculations, with speed and accuracy. Yet, today’s computers are not as good at performing tasks such as abstract reasoning, concept formulation, and strategic planning (Yao et al., 2018).

What Accountants Need to Know About AI

Artificial intelligence is divided into a variety of sub-fields, including but not limited to machine reasoning, machine learning, deep learning, and natural language processing. Each of these sub-fields has essential applications in accounting. In effect, AI has fields that may even be nested as sub-fields, such as Deep Learning (DL) is a sub-field of Machine Learning (ML), as illustrated in Figure 1.1.

Artificial Intelligence

As stated in the introduction, artificial intelligence is defined broadly as a computer program or software application that can imitate or simulate human behavior. This definition works well with a variety of technologies that we use every day. For example, the words that you are currently reading were spoken into a dictation tool in a popular word processing software. This form of AI uses speech recognition software to translate audio data into text, mimicking the actions of a human transcriber.

However, it is important to recognize that a generally accepted definition of artificial intelligence remains open for debate (Dobrev, 2012). An influential textbook in AI research entitled *Artificial Intelligence: A Modern Approach* offers four possibilities: systems that think like humans, systems that act like humans, systems that think rationally, and systems that act rationally (Russell & Norvig, 2010).

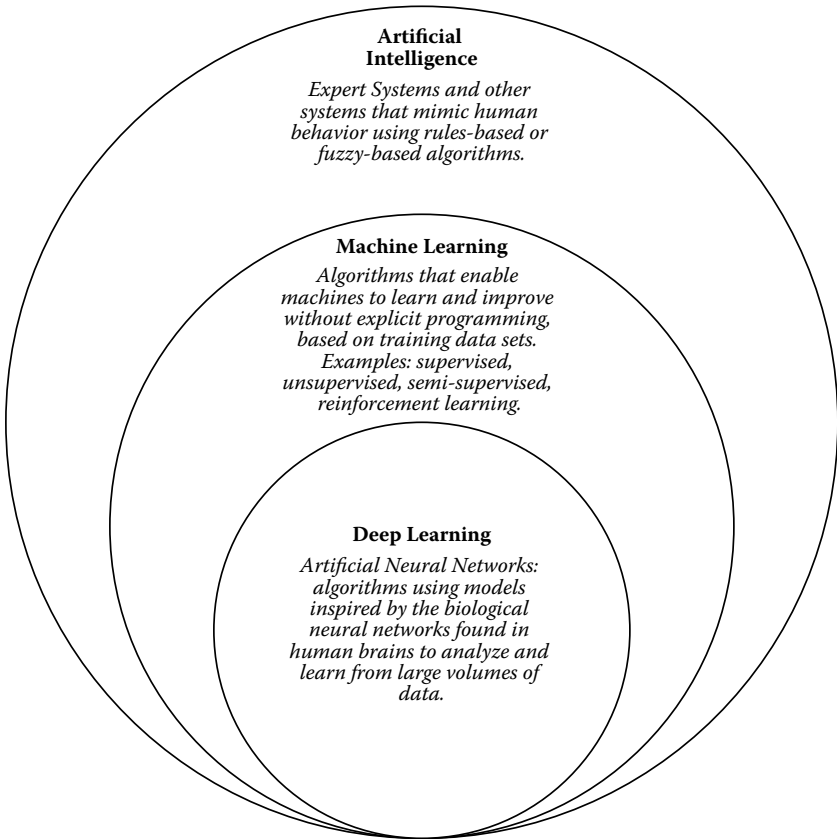


Figure 1.1 Relationship between AI, ML, and DL.

Today speech recognition is considered a form of AI and tomorrow it may be relegated to straightforward automation. The bottom line here is that what constitutes AI continues to evolve with time and technological advances. Although a consensus may not exist on a formal definition of AI, experts often make a distinction between two sub-types of AI – artificial narrow intelligence (ANI) and artificial general intelligence (AGI).

Artificial Narrow Intelligence (ANI)

Artificial narrow intelligence (ANI), also known as weak or narrow AI, focuses on a task, such as speech recognition, computers that can play

6 *What Accountants Need to Know*

chess, or autonomous vehicles. Virtually all of the AI with which we are familiar can be classified as ANI, as computer scientists have yet to create machines that can experience emotion and the ability to perceive or feel things (Jajal, 2018). ANI machines that are programmed for these specific tasks are typically much better than their humans at doing the same task.

Examples of narrow AI include chatbots that are built for the narrow purpose of conversation. Chatbots are capable of verbal responses, using algorithms and a database, based on a statement or question.

Artificial General Intelligence (AGI)

The goal of **artificial general intelligence (AGI)**, also known as strong or broad AI, is to create machines capable of performing all the cognitive tasks of the human brain. The essential elements of AGI include: (1) the ability to apply knowledge from one domain to another; (2) the ability to plan for the future based on experience and knowledge; and (3) the ability to adapt based on changing circumstances (Walch, 2019).

Machine Reasoning

Machine reasoning (MR) is the ability of a computer to draw conclusions from a knowledge base using automated inference techniques that can imitate or simulate human inference, such as deduction and induction. León Bottou, a reasoning expert, provides a more technical definition of MR as an “algebraic manipulation of previously acquired knowledge in order to answer a new question” (Bottou, 2014, p. 136).

Expert Systems

The initial applications of reasoning systems were referred to as expert systems (ES). **Expert systems** are an early form of AI, first developed in the 1970s. They are computer systems that store knowledge from human experts to emulate human decision-making. To become an expert, one must have deep knowledge of both facts and rules as well as practical experience in a particular domain (Negnevitsky, 2011). Such expertise can be gained through formal training or hands-on experience. ES works by making recommendations or drawing conclusions on a narrow topic using a rules-based engine applied against a database. ES is used in a variety of disciplines in accounting, such as

audit, tax, management accounting, and personal financial planning (Yang & Vasarhelyi, 1993).

Expert systems can be classified as rules-based or fuzzy-based. Rules-based expert systems employ a set of detailed instructions, called **algorithms**, that apply rules to interpret how to react to various scenarios. For example, a tax software program may use the following rule: if the adjusted gross income equals X dollars, then the applicable federal tax is equal to Y dollars.

Fuzzy-based expert systems use fuzzy logic – a logical system in which conditions cannot be described in binary terms such as true or false, 0 or 1. Fuzzy logic is useful in situations that involve imprecision and vagueness (Negnevitsky, 2011). For example, an algorithm using fuzzy logic was used in an expert system designed to assist with fraud detection of settled insurance claims (Pathak et al., 2005). These researchers also suggest that fuzzy logic expert systems could be used by auditors to assess control risks and detection risks on an audit engagement.

Today's machine reasoning systems go beyond the rules-based capabilities of ES. AI-enabled MR systems can now learn using machine learning algorithms and artificial neural networks to draw conclusions through data analysis. As a result, there are vast opportunities for accountants to utilize MR systems to enhance productivity.

Machine Learning

Machine learning (ML) is a subset of AI (as illustrated in Figure 1.1) that uses algorithms to analyze data to carry out specific tasks, such as making predictions, without relying on explicit programming as in rule-based expert systems. Machine learning uses pattern recognition and inference to learn from data. The larger the data set, the more examples from which the algorithm can learn through trial and error (CPA Canada, & AICPA, 2019). Machine learning is used by Netflix to make recommendations for TV shows and movies based on what the user previously viewed, thereby enhancing the user experience. The more content that a machine learning AI views, the more refined and accurate its predictions will be. To cite another example, Amazon uses machine learning to analyze purchasing data on its products to forecast demand, identify fraudulent purchases, and provide customized recommendations and promotions (Camhi & Pandolh, 2017).

We have so far discussed machine learning in consumer-driven fields, but what does it have to offer the field of accounting?

Since machine learning enables systems to be adaptive (Negnevitsky, 2011), computers equipped with ML can learn from experience, resulting in improved performance over time. As an example, KPMG³ employs IBM's machine learning algorithms to assist their clients with compliance with the IFRS 16 lease accounting standard (Samuel, 2018). Specifically, the KPMG Contract Abstraction Tool extracts data from tens of thousands of contracts, each of which may be hundreds of pages long, and analyzes the data for compliance with the leasing standard. As the program examines more contracts, it becomes "smarter" and can offer more refined and targeted feedback. Markus Kreher, Global Head of Accounting Advisory Services and Head of Finance Advisory, KPMG in Germany, said: "We've effectively trained the solution to read and understand contracts just like an attorney would" (Samuel, 2018, para. 4).

Similarly, Deloitte uses an award-winning machine learning tool called Argus to review, identify, and extract key accounting information from any type of electronic document. Argus learns from every human interaction and can analyze sales and leasing contracts, employment agreements, invoices, meeting minutes, financial statements, and legal letters (Deloitte, 2015), eliminating a time-consuming and costly manual task normally performed by auditors. Thus, Argus enables Deloitte auditors to be more productive by spending their time interpreting results and exercising professional skepticism.

Possible applications for ML in accounting include computers that learn from audit failures by analyzing data sets that were the basis for incorrectly issuing a clean audit opinion. Another application of ML could be on tax positions taken by firms that were rejected by the federal and state tax authorities.

There are four main types of machine learning: supervised, unsupervised, semi-supervised, and reinforced.

Supervised Learning

Supervised learning is a process in which a computer algorithm learns from a set of training data that is labeled (tagged with certain attributes such as poor credit or good credit) and paired as input and output variables (X results in Y). Supervised learning is often used for classification (e.g., identifying fraudulent and non-fraudulent transactions) and prediction (e.g., predicting the number of uncollectible accounts based on the aging of individual customers). For example, suppose an auditor wants to predict the number of uncollectible accounts based on the aging of customer balances.

A developer would provide the model with historical data that includes the input variable (aging of individual customers) and the output data (amount of uncollectible accounts written off). A supervised ML model would then “learn” the rules based on the historical data and then predict the amount of uncollectible accounts when new aging data is provided.

Unsupervised Learning

Unsupervised learning, on the other hand, is a process in which the computer algorithm only learns from unlabeled input data (i.e., the outputs are unknown). For example, consider a controller of an accounting department who wants to better understand why customers do not pay within 30 days. An unsupervised machine learning model would teach the algorithm to identify trends and patterns related to the input data. The algorithm might identify that customers with outstanding accounts receivable balances over 120 days might have similar zip codes, income levels, or credit scores. One of the most common approaches for unsupervised learning is cluster analysis, whereby input data is grouped based on similarities.

Semi-supervised Learning

Semi-supervised learning is a hybrid of supervised and unsupervised learning using both labeled and unlabeled data during the training process. As many real-world data sets often contain incorrect or missing labels (Yao et al., 2018), using a semi-supervised approach could lead to improved learning compared to a purely supervised or unsupervised method.

Reinforcement Learning

Reinforcement learning is a process in which a computer algorithm trains itself, learning from data through trial and error. The agent is rewarded for performing specific tasks correctly and penalized for performing tasks incorrectly. These algorithms are designed to maximize rewards and minimize penalties.

Deep Learning

Deep learning (DL) is a subset of machine learning (as illustrated in Figure 1.1) that uses artificial neural networks to discover patterns

from data. **Artificial neural networks (ANNs)** are a collection of artificial neurons (also known as units) that receive data as an input, and then logic is applied to produce an output. ANNs are frequently used for pattern recognition such as facial recognition in pictures and videos, speech recognition, social networks, etc., frequently accelerated by tensor processing. A **tensor processing unit (TPU)** is an application-specific circuit developed by Google that is used to increase processing speeds for machine learning and deep learning applications. Whereas traditional machine learning relies heavily on the guidance provided by human programmers, DL models use ANNs (inspired by the biological neural networks found in human brains) to analyze large volumes of data to teach itself.

Deep learning can be used in a variety of applications, such as facial recognition, image classification, speech recognition, and text translation. Current applications of DL include self-driving vehicles that slow down as it approaches a pedestrian cross-walk, ATM machines that reject a counterfeit banknote, and a smartphone app that translates the image of a street sign in a foreign language (The Mathworks & MATLAB, 2018).

EY uses deep learning technology to reconstruct documents (e.g., invoices and contracts used during an audit) that were poorly-scanned using optical recognition software (Duffy, 2019). Researcher Sophia Sun (2019) promotes the use of DL to support audit decision making in two key areas: information identification and judgment support. DL algorithms can analyze semi-structured data (e.g., text data from supporting documents such as invoices, contracts, etc.) and unstructured data (such as images, audio, and video from the internet, social media, etc.), which will aid in information identification. Further, Sun notes that the predictive performance of DL models is superior to classic machine learning when the volume of data and the number of variables is large. To maximize the use of DL in an audit context, Sun suggests designing and developing a data warehouse to train the algorithms to make predictions and assist with audit procedures.

Natural Language Processing

Natural language processing (NLP) is a subfield of AI that focuses on the interaction of computers and people using human languages. The Siri application on Apple's iPhone and Amazon's Alexa uses NLP to understand requests for completing a task.

NLP can be broken down into two types: natural language understanding and natural language generation.

Natural language understanding (NLU) enables computers to understand instructions provided in human language. For example, when a user asks their iPhone for weather conditions by using the following voice command, “Hey Siri, what’s the latest weather?” Siri understands their command and retrieves the information by searching on the Internet.

Natural language generation (NLG) enables computers to produce human language so that people can understand computers. After Siri performed a search of the latest weather, it replies with an answer such as the following example: “it’s currently clear and 28 degrees in Philadelphia. Expect partly cloudy skies starting tonight. Today’s high will be 33 degrees, and the low will be 23 degrees”. Siri understood that the user was asking it to look up the weather conditions at a specific location and time, and then returned the results in spoken English words. NLG can also convert data visualizations, such as charts and graphs, into a verbal description that can be understood by humans (CPA Canada & AICPA, 2019).

The potential for using NLP in accounting is substantial, as it can be used to process and analyze large amounts of data. NLP has been used to analyze various textual documents related to corporate financial performance and compliance with accounting standards and regulations (Fisher et al., 2016). Text mining uses NLP to extract meaning from textual data. Applications of NLP include text mining components such as semantics analysis, text classification, text summarization, and text translation. Additional applications include speech recognition, question-answering systems, and chatbots.

Data Mining

Data mining is a process of analyzing large data sets (i.e., “big data”) to discover previously unknown patterns or relationships. Data mining tools are typically applied to structured data, that is, highly formatted data contained in relational databases such as in an ERP system and, therefore, can be very helpful for accountants.

As illustrated in Figure 1.2, data mining intersects with but is not considered a subfield of AI. As computer scientist Xavier Amatriain notes, “machine learning can be used for data. However, data mining can use other techniques on top of machine learning” (Amatriain, 2016, para. 5).

12 *What Accountants Need to Know*

Like machine learning, data mining relies on reading large amounts of data to discover patterns and arrive at conclusions. However, technology expert Bernard Marr identified notable differences between data mining and machine learning. First, data mining looks for patterns that already exist in historical data, whereas ML attempts to predict future outcomes based on the given data. Second, the rules or patterns are unknown at the beginning of the data mining process, whereas the rules are programmed into the computer to understand the data with machine learning. Third, data mining relies on human intervention throughout the process, whereas much of the learning with ML is automatic. Finally, data mining uses an existing data set, such as a data warehouse, whereas ML learns from a training data set and then makes predictions using new data sets (Marr, n.d.).

By using data mining techniques on general ledger transactions, accountants can potentially unlock valuable insights and improve decision-making. For example, auditors can use data mining to help detect fraudulent purchases or outlier transactions that require further investigation. Management accountants can use data mining to assess the financial risk of a business entity, such as “trading partners, corporate affiliates, investment partners, and takeover targets” (Calderon et al., 2003, p. 7).

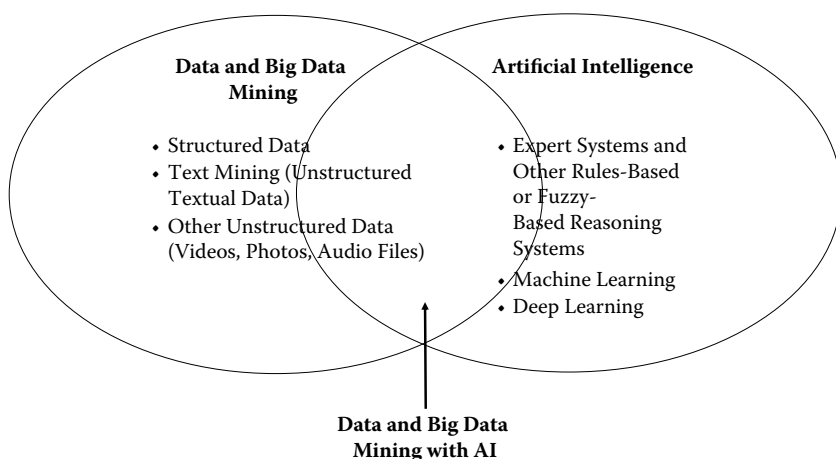


Figure 1.2 Relationship between (big) data mining and AI.

Text Mining

Text mining is a process of extracting information from various text sources (such as Word documents, PDF files, social media posts, emails, websites, articles, XML files, and others) to discover patterns, trends, and themes. The text found in these documents is typically unstructured, that is, they are not in a predefined format that can be analyzed through data analytics software such as IDEA or ACL. Text mining is performed in two steps: 1) imposing structure on the text data sources and then 2) using data mining techniques to extract relevant information (Sharda et al., 2014).

Text mining is useful in fields that have a large amount of textual data, as is the case with accounting. For example, auditors review contracts, invoices, legal letters, SEC filings, earnings, conference calls, news articles, and much more.

Professor Aldhizer (2017) of Wake Forest University suggests that forensic and audit practices consider using text analytics for high-risk engagements. Aldhizer notes that text analytics could be used for concept extraction to identify incriminating words from social media posts and emails. Text mining could be used to ensure proper reporting under lease and revenue recognition (Iowa State University, n.d.).

Robotic Process Automation (RPA) and AI

Although RPA is not considered a form of AI by some experts (CPA Canada & AICPA, 2019), this technology enables machines to perform functions usually carried out by humans. Firms are using RPA, coupled with AI capabilities, to increase productivity and performance. Thus, accountants need to understand what RPA is and how it can be combined with AI technologies. **Robotic process automation (RPA)** is a software application (robot or bot) that automates a business process by replicating the actions of humans performing tasks within digital systems, such as manipulating or transferring data. In many cases, the automation involves minimal coding and is typically activated by using a smart screen recording. Keep in mind that these robots are not physical steel machines found in the old science-fiction movies or television shows. RPA lends itself well to high-volume and repeatable tasks traditionally performed by humans. For example, RPA can be used to process bulk transactions, such as processing vendor invoices for payment.

By using RPA, businesses can achieve increased productivity and accuracy at a lower cost. The exponential growth in RPA

adoption has been facilitated by the availability of software providers such as UiPath, Automation Anywhere, and BluePrism. Many RPA providers will allow individuals to have a limited version of their software to begin building bots. Examples of RPA in accounting include performing bank reconciliations, automating cash applications, and tracking accounts payable.

Application Programming Interfaces (API) and AI

An **application programming interface (API)** is a set of functions or rules that facilitates communication between applications, databases, and devices. An analogy to understand how an API works is a restaurant waiter who takes an order relays it to the kitchen and then returns the prepared food to the customer. APIs are like digital messengers – they allow the weather app on your phone to transmit and retrieve information from a third-party app such as the Weather Channel. As another example, APIs enable customers to purchase movie tickets through a site like Fandango by signing into their Facebook or Google accounts.

APIs are used extensively by AI and ML programmers. For example, Google's Prediction API provides cloud-based access to machine learning, natural language processing, recommendation engines, pattern recognition, and prediction services (Rizk, 2018). In a blog post entitled "Top 20 APIs You Should Know in AI and Machine Learning," Oleksii Kharkovyna (2019) notes that his preferred APIs are most suitable for AI and ML purposes. BigML, for example, uses APIs to create data sets, models, and predictions that use ML technology. As another example, Anaconda Enterprise is a data science platform that uses APIs to deploy machine learning models.

Best Programming Languages Accountants Should Learn for Artificial Intelligence Applications

Accountants with a strong technical background in programming languages will be best positioned to work with data scientists in implementing AI initiatives. For those do-it-yourself accountants who may want to customize AI technologies to solve applications, learning a programming language will be quite useful. Proficiency with popular open-source programming tools for AI, such as Python and R, will enable accountants to customize AI applications to suit the unique business needs of their firms or clients. Other tools, such as SQL and

NoSQL, will be useful for retrieving, editing, and manipulating data to be used with AI applications.

Accountants without a strong background in programming languages can also help develop AI-enabled applications by using “no-code” and “low-code” platforms. These platforms use drag-and-drop editors for application development rather than traditional coding languages. Some of the world’s leading technology companies offer these solutions. For example, the “Create ML” application by Apple allows users to develop and train ML models using different datasets. Google’s AI Platform supports “Kubeflow”, their open-source application that enables developers to build AI/ML projects. Lastly, Microsoft’s Cognitive Toolkit “CNTK” enables applications built for the Azure platform and allows users to develop customized applications for computer vision, natural language understanding, facial recognition, and prediction models. No-code and low-code tools are undoubtedly useful for accountants in implementing AI initiatives.

Notes

- 1 As used throughout this book, “KMPG” refers to KPMG International Cooperative (“KPMG International”). Member firms of the KPMG network of independent firms are affiliated with KPMG International. KPMG International provides no client services.
- 2 As used throughout this book, “Deloitte” means Deloitte & Touche LLP, a subsidiary of Deloitte LLP. Please see www.deloitte.com/us/about for a detailed description of the legal structure. Certain services may not be available to attest clients under the rules and regulations of public accounting.
- 3 As used throughout this book, “KMPG” refers to KPMG International Cooperative (“KPMG International”). Member firms of the KPMG network of independent firms are affiliated with KPMG International. KPMG International provides no client services.

References

- Aldhizer, G. R. (2017). Visual and text analytics. *The CPA Journal*, 87(6), 30–33. <https://www.cpajournal.com/2017/06/20/visual-text-analytics/>.
- Amatriain, X. (2016, January 14). *What’s the relationship between machine learning and data mining?* Medium. <https://medium.com/@xamat/what-s-the-relationship-between-machine-learning-and-data-mining-8c8675966615>.
- Bottou, L. (2014). From machine learning to machine reasoning: An essay. *Machine Learning*, 94(2), 133–149. <https://doi.org/10.1007/s10994-013-5335-x>.
- Bringsjord, S., & Govindarajulu, N. S. (2018). Artificial intelligence. *The Stanford Encyclopedia of Philosophy* (Summer 2020 Edition). Edward N. Zalta. <https://plato.stanford.edu/entries/artificial-intelligence/#HistAI>.

- Burns, M. B., & Igou, A. (2019). "Alexa, write an audit opinion": Adopting intelligent virtual assistants in accounting workplaces. *Journal of Emerging Technologies in Accounting*, 16(1), 81–92. <https://doi.org/10.2308/jeta-52424>.
- Calderon, T. G., Cleh, J. J., & Kim, I. W. (2003). How large corporations use data mining to create value. *Management Accounting Quarterly*, 3(4), 1–12.
- Camhi, J., & Pandolh, S. (2017, April 17). *Machine learning driving innovation at Amazon*. Business Insider. <https://www.businessinsider.com/machine-learning-driving-innovation-at-amazon-2017-4>.
- Colom , R, Karama , S, Jung , RE, & Haier , RJ (2010). Human intelligence and brain networks. *Dialogues Clin Neurosci*, 12(4), 489–501.
- CPA Canada, & AICPA. (2019). *A CPA' s Introduction to AI: From Algorithms to Deep Learning, What You Need to Know*. CPA Canada. <https://www.cpacanada.ca/en/business-and-accounting-resources/other-general-business-topics/information-management-and-technology/publications/a-cpa-introduction-to-ai>.
- Deloitte. (2015). *Deloitte wins 'Audit Innovation of the Year' at 2015 International Accounting Bulletin awards*. <https://www2.deloitte.com/ch/en/pages/audit/articles/deloitte-wins-iab-audit-innovation-award.html>.
- Dobrev, D. (2012). A definition of artificial intelligence. *Mathematica Balkanica, New Series*, 19, 1–7.
- Duffy, N. (2019, August 8). *How AI is transforming business right now*. EY. https://www.ey.com/en_us/innovation/how-ai-is-transforming-business-right-now.
- Fisher, I. E., Garnsey, M. R., & Hughes, M. E. (2016). Natural language processing in accounting, auditing, and finance: A synthesis of the literature with a roadmap for future research. *Intelligent Systems in Accounting, Finance, and Management*, 23(3), 157–214. <https://doi.org/10.1002/isaf.1386>.
- ICAEW. (2017). *Artificial intelligence and the future of accountancy*. <https://www.icaew.com/-/media/corporate/files/technical/information-technology/thought-leadership/artificial-intelligence-report.ashx>.
- Iowa State University. (n.d.). *Textual Analytics for Accounting and Auditing*. <https://www.ivybusiness.iastate.edu/files/2018/12/Janvrin-Textual-Analysis-Presentation-Dec-14-2018.pdf>.
- Jajal, T. (2018, May 21). *Distinguishing between Narrow AI, General AI, and Super AI*. Medium. <https://medium.com/@tjajal/distinguishing-between-narrow-ai-general-ai-and-super-ai-a4bc44172e22>.
- Kharkovyna, O. (2019, March 11). *Top 20 APIs you should know in AI and Machine Learning*. Towards Data Science. <https://towardsdatascience.com/top-20-apis-you-should-know-in-ai-and-machine-learning-8e08515198b3>.
- Marr, B. (n.d.). What is the difference between data mining and machine learning? Bernard Marr & Co. <https://bernardmarr.com/default.asp?contentID=1741#:~:text=Data%20mining%20is%20used%20on,predictions%20about%20new%20data%20sets>.

- Merriam-Webster. (2020). Intelligence. In *Merriam-Webster.com dictionary*. <https://www.merriam-webster.com/dictionary/intelligence>.
- Negnevitsky, M. (2011). *Artificial Intelligence: A guide to intelligent systems* (3rd ed.). Addison-Wesley, an imprint of Pearson.
- New York State Society of CPAs. (2015). Technology in accounting history. *CPA Journal*, 85(11), 16–17. <https://www.cpajournal.com/2017/11/30/technology-accounting-history/>.
- Pathak, J., Vidyarthi, N., & Summers, S. L. (2005). A fuzzy-based algorithm for auditors to detect elements of fraud in settled insurance claims. *Managerial Auditing Journal*, 20(6), 632–644. <https://doi.org/10.1108/02686900510606119>.
- Pepe, A. A. (2011, April 19). *The evolution of technology for the accounting profession*. CPA Practice Advisor. <http://www.cpapracticeadvisor.com/article/10263076/the-evolution-of-technology-for-the-accounting-profession>.
- Rizk, A. (2018, January 23). *How API and artificial intelligence can be complementary?* API Friends. <https://apifriends.com/api-management/api-and-artificial-intelligence/>.
- Russell, S., & Norvig, P. (2010). *Artificial intelligence – A modern approach*. Prentice-Hall.
- Samuel, J. (2018, March 21). *KPMG applying IBM artificial intelligence to help businesses efficiently meet IFRS 16 lease account requirements*. KPMG. <https://home.kpmg/xx/en/home/media/press-releases/2018/03/kpmg-applying-ibm-ai-to-help-businesses-meet-ifrs-16.html>.
- Schwab, K. (2017). *The fourth industrial revolution*. Crown Business.
- Sharda, R., Delen, D., & Turban, E. (2014). *Business intelligence* (3rd ed.). Pearson.
- Siegel, J., Shim, J., Walker, J., Qureshi, A., O'Callaghan, S., & Koku, P. (2003). *The artificial intelligence handbook: Business applications in accounting, banking, finance, management, and marketing*. South-Western, a division of Thomson Learning.
- Sun, T. S. (2019). Applying deep learning to audit procedures: An illustrative framework. *Accounting Horizons*, 33(3), 89–109. <https://doi.org/10.2308/acch-52455>.
- The Mathworks & MATLAB. (2018). *Introducing Deep Learning with MATLAB: What is Deep Learning?* https://www.mathworks.com/content/dam/mathworks/ebook/gated/80879v00_Deep_Learning_ebook.pdf.
- Untapped Editorial Team. (2018, August 29). *The history of accounting – From record keeping to artificial intelligence discover how the history of accounting evolved*. Medius. <https://www.mediusflow.com/en/untapped/articles/people/history-of-accounting>.
- Vien, C. (Host). (2018, July 30). *Using drones to enhance audits* [Audio Podcast]. *Journal of Accountancy*. <https://www.journalofaccountancy.com/podcast/using-drones-to-enhance-audits.html>.
- Walch, K. (2019, October 4). *Rethinking weak vs. strong AI*. Forbes. <https://www.forbes.com/sites/cognitiveworld/2019/10/04/rethinking-weak-vs-strong-ai/#4cf807ab6da3>.

- Weiner, I. B., & Freedheim, D. K. (2013). *Handbook of psychology, history of psychology* (2nd ed.). John Wiley & Sons.
- Wittbold, K. A., Carroll, C., Iansiti, M., Zhang, H. M., & Landman, A. B. (2020). *How hospitals are using AI to Battle Covid-19*. Harvard Business Review. <https://hbr.org/2020/04/how-hospitals-are-using-ai-to-battle-covid-19>.
- Yang, D. C., & Vasarhelyi, M. (1993). The application of expert systems in libraries. *The Journal of Academic Librarianship*, 19(1), 45–46. [https://doi.org/10.1016/0099-1333\(93\)90774-y](https://doi.org/10.1016/0099-1333(93)90774-y).
- Yao, M., Jia, M., & Zhou, A. (2018). *Applied artificial intelligence: A handbook for business leaders* (N. Zhang, Ed.). TOPBOTS Inc.